



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to CHENNAMMAL AGENCIES

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

#### Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



**Route Summary:**  
**Total Distance: 156.18 km**  
**Estimated Duration: 3.5 hours**  
**Adjusted Duration (Heavy Vehicle): 4.4 hours**



Start: (11.4381, 77.8734)  
End: (12.362, 78.541)



## Welcome to the Journey Risk Management Study

### 1. Overview of the Route Map

This route begins at CVQF+23W, Sangagiri, Tamil Nadu, progresses through Bhavani Main Road in Padaiveedu, and then passes through Coimbatore Rd in Kondalampatti, Salem, finally reaching 9G6R+Q95 in Kallavi, Tamil Nadu. The route stretches approximately 156.18 kilometers, characterized by a mix of urban and rural landscapes. The journey typically takes about 3.51 hours when accounting for the transport of heavy vehicles carrying hazardous materials.

### 2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate with distinct wet and dry seasons. The monsoon season, lasting from June to September, can bring heavy rainfall, potentially causing flooding and slippery road surfaces. During summer months (March to May), high temperatures may lead to road surface issues such as softening asphalt, which can affect vehicle handling. Additionally, occasional dense fog can occur in the cooler months of December and January, reducing visibility significantly.

### 3. Analysis of Traffic Patterns

- Sangagiri and Salem:** Moderate to heavy traffic, particularly during rush hours (8-10 AM and 5-8 PM). Salem, in particular, may have congestion near market areas and industrial zones.
- Rural Segments:** Lower traffic density, but agricultural vehicles and local commuters can cause slowdowns.
- Congestion-Prone Areas:** Key congestion points include the towns of Sangagiri and Kondalampatti, especially where roads intersect with local businesses and market areas.

## 4. Assessment of Road Quality and Infrastructure

- **Road Quality:** The route encompasses both well-maintained highways and smaller rural roads, which may have potholes or uneven surfaces.
- **Infrastructure:** Expect standard two-lane highways in rural sections and broader roads in urban areas, but watch for narrow sections near villages and potential obstructions from market activity.

## 5. Suggestions for Alternative Routes for Emergencies

- **Alternate Route A:** A potential diversion could bypass Salem via State Highway 188, though this may add both time and distance.
- **Alternate Route B:** Consider FR48+7XQ to FR54, leading to Kallavi by avoiding some congested urban centers.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

Transport of hazardous materials in Tamil Nadu requires adherence to strict regulations, including obtaining necessary permits, using appropriate safety markings, and operating during designated times (usually avoiding nighttime transport unless specifically permitted).

## 7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

While data on specific incidents is limited, the region has witnessed some accidents involving heavy vehicles, primarily due to driver fatigue and adverse weather conditions. Notably, intersections and sharp bends in rural areas have been identified as potential risk points.

## 8. Environmental Considerations and Sensitive Areas

The route crosses near agricultural regions and small villages, necessitating caution to avoid spillages that could affect local water supplies or agricultural fields. Proximity to wildlife reserves should also prompt adherence to speed limits to prevent disturbances.

## 9. Analysis of Communication Coverage

Mobile networks generally provide good coverage along the route, especially near urban areas. However, certain rural sections may experience intermittent signal loss, particularly around hilly or forested terrain.

## 10. Estimated Emergency Response Times for Different Route Segments

- **Urban Areas (Salem, Sangagiri):** Emergency services response times can be estimated at approximately 15-30 minutes due to better infrastructure and proximity to service facilities.
- **Rural Sections:** Response times may extend to 45-60 minutes due to access issues and the distance from emergency service hubs.

## 12. Overall Summary of Risk Assessment



The route poses moderate risk levels chiefly due to variable road conditions, potential for traffic congestion, and environmental factors such as weather. Key risk mitigation steps include thorough route planning to avoid peak hours and preparing for adverse weather; ensuring vehicle maintenance to handle variable road surfaces; and maintaining communication with local authorities and emergency services. Regular checks and adherence to regulations regarding hazardous materials will further ensure a safer transit.

Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Roundabout | High       | 11.63266, 78.12699 | 15 KM/Hr    |
| 1  | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 2  | Turn       | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 3  | Turn       | High       | 11.44900, 77.87406 | 15 KM/Hr    |
| 4  | Turn       | High       | 11.45353, 77.85697 | 15 KM/Hr    |
| 5  | Turn       | Medium     | 11.45781, 77.85203 | 30 KM/Hr    |
| 6  | Turn       | High       | 11.46307, 77.84941 | 15 KM/Hr    |
| 7  | Turn       | Medium     | 11.45546, 77.81800 | 30 KM/Hr    |
| 8  | Blind Spot | Blind Spot | 11.45605, 77.81683 | 10 KM/Hr    |
| 9  | Turn       | High       | 11.53573, 77.95794 | 15 KM/Hr    |
| 10 | Turn       | Medium     | 11.62100, 78.10723 | 30 KM/Hr    |
| 11 | Blind Spot | Blind Spot | 11.62130, 78.10776 | 10 KM/Hr    |
| 12 | Blind Spot | Blind Spot | 11.61235, 78.10438 | 10 KM/Hr    |
| 13 | Turn       | Medium     | 11.61235, 78.10495 | 30 KM/Hr    |
| 14 | Blind Spot | Blind Spot | 11.62871, 78.12299 | 10 KM/Hr    |
| 15 | Blind Spot | Blind Spot | 11.62788, 78.12219 | 10 KM/Hr    |
| 16 | Turn       | High       | 11.63302, 78.12742 | 15 KM/Hr    |
| 17 | Turn       | Medium     | 11.62938, 78.16601 | 30 KM/Hr    |
| 18 | Turn       | Medium     | 11.65825, 78.20469 | 30 KM/Hr    |
| 19 | Turn       | Medium     | 11.66366, 78.20649 | 30 KM/Hr    |
| 20 | Turn       | High       | 11.67116, 78.23102 | 15 KM/Hr    |
| 21 | Turn       | Medium     | 11.67182, 78.23141 | 30 KM/Hr    |
| 22 | Turn       | Medium     | 11.67500, 78.23962 | 30 KM/Hr    |
| 23 | Turn       | Medium     | 11.80474, 78.33353 | 30 KM/Hr    |
| 24 | Turn       | Medium     | 11.81272, 78.34092 | 30 KM/Hr    |
| 25 | Turn       | Medium     | 11.83514, 78.35305 | 30 KM/Hr    |
| 26 | Turn       | Medium     | 11.99608, 78.44589 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 12.04286, 78.48352 | 30 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 28 | Turn      | Medium     | 12.04845, 78.48385 | 30 KM/Hr    |
| 29 | Turn      | Medium     | 12.04958, 78.48314 | 30 KM/Hr    |
| 30 | Turn      | Medium     | 12.06149, 78.47722 | 30 KM/Hr    |
| 31 | Turn      | Medium     | 12.06932, 78.47655 | 30 KM/Hr    |
| 32 | Turn      | Medium     | 12.07112, 78.47724 | 30 KM/Hr    |
| 33 | Turn      | Medium     | 12.11669, 78.49766 | 30 KM/Hr    |
| 34 | Turn      | Medium     | 12.17764, 78.54800 | 30 KM/Hr    |
| 35 | Turn      | Medium     | 12.17847, 78.54856 | 30 KM/Hr    |
| 36 | Turn      | Medium     | 12.18223, 78.54843 | 30 KM/Hr    |
| 37 | Turn      | High       | 12.18614, 78.54962 | 15 KM/Hr    |
| 38 | Turn      | Medium     | 12.20953, 78.54358 | 30 KM/Hr    |
| 39 | Turn      | High       | 12.21030, 78.54402 | 15 KM/Hr    |
| 40 | Turn      | Medium     | 12.25956, 78.53301 | 30 KM/Hr    |
| 41 | Turn      | Medium     | 12.26092, 78.53324 | 30 KM/Hr    |
| 42 | Turn      | Medium     | 12.26207, 78.53436 | 30 KM/Hr    |
| 43 | Turn      | Medium     | 12.31095, 78.53818 | 30 KM/Hr    |
| 44 | Turn      | High       | 12.36356, 78.54371 | 15 KM/Hr    |
| 45 | Turn      | High       | 12.36375, 78.54324 | 15 KM/Hr    |

## Emergency Locations

|    | type     | name                                                            | coordinates            | speed_limit | risk_level |
|----|----------|-----------------------------------------------------------------|------------------------|-------------|------------|
| 2  | hospital | C. N. Hospital                                                  | 11.55724, 78.0096      | 30 km/h     | Medium     |
| 3  | hospital | Vinayaka mission Kirubananadah variyar medical college hospital | 11.584984, 78.0618417  | 30 km/h     | Medium     |
| 4  | hospital | vims                                                            | 11.5854239, 78.0639188 | 30 km/h     | Medium     |
| 7  | hospital | Vinayaka Mission’s Kirupananda Variyar Medical College          | 11.5977669, 78.0850992 | 30 km/h     | Medium     |
| 11 | hospital | Thayammal Hospitals                                             | 11.6237472, 78.1177631 | 30 km/h     | Medium     |
| 12 | hospital | Vanitha Hospital                                                | 11.6239931, 78.1184622 | 30 km/h     | Medium     |
| 14 | hospital | Universal Cancer Hospital                                       | 11.6336739, 78.1267767 | 30 km/h     | Medium     |
| 15 | hospital | Priyam Speciality Hospital                                      | 11.63446, 78.124388    | 30 km/h     | Medium     |

|    | type     | name                               | coordinates               | speed_limit | risk_level |
|----|----------|------------------------------------|---------------------------|-------------|------------|
| 16 | hospital | AVM Hospital                       | 11.6264369,<br>78.1394204 | 30 km/h     | Medium     |
| 17 | pharmacy | மருந்தகம்                          | 11.6251501,<br>78.1470098 | 30 km/h     | Medium     |
| 18 | hospital | Rps Hospital                       | 11.6260528,<br>78.1474921 | 30 km/h     | Medium     |
| 19 | hospital | மருத்துவமனை நிலம்                  | 11.625344,<br>78.146058   | 30 km/h     | Medium     |
| 20 | hospital | Dharan Hospital                    | 11.6252302,<br>78.1451753 | 30 km/h     | Medium     |
| 21 | hospital | Akitha Hospital                    | 11.6256027,<br>78.1489875 | 30 km/h     | Medium     |
| 22 | hospital | Deepam Speciality Hospital         | 11.6269734,<br>78.1476357 | 30 km/h     | Medium     |
| 23 | hospital | Udhayam Hospital                   | 11.625819,<br>78.145958   | 30 km/h     | Medium     |
| 24 | hospital | Manima Clinic                      | 11.623248,<br>78.1465721  | 30 km/h     | Medium     |
| 26 | hospital | Dr.Thiru Neuro Multi Speciality    | 11.6623376,<br>78.203353  | 30 km/h     | Medium     |
| 28 | hospital | M. Thathanur Primary Health Center | 11.6824017,<br>78.2445813 | 30 km/h     | Medium     |
| 29 | hospital | Goverment Hospital                 | 11.7580777,<br>78.2969128 | 30 km/h     | Medium     |
| 32 | hospital | Government Hospital Chinnankuppam  | 12.0110294,<br>78.4553224 | 30 km/h     | Medium     |
| 37 | hospital | Nanjaya Hospital                   | 12.0550888,<br>78.4821929 | 30 km/h     | Medium     |
| 38 | hospital | Subam Hospital                     | 12.0595128,<br>78.4766713 | 30 km/h     | Medium     |
| 39 | hospital | Government Hospital, Harur         | 12.061178,<br>78.4756913  | 30 km/h     | Medium     |
| 40 | hospital | Government Hospital, Uthangarai    | 12.26612,<br>78.535949    | 30 km/h     | Medium     |

Crowded Spots

|   | type   | name                       | coordinates               | speed_limit | risk_level |
|---|--------|----------------------------|---------------------------|-------------|------------|
| 0 | school | KRP Matric. Hr. Sec School | 11.4546193,<br>77.8142445 | 30 km/h     | Medium     |

|    | type       | name                                                    | coordinates            | speed_limit | risk_level |
|----|------------|---------------------------------------------------------|------------------------|-------------|------------|
| 5  | school     | Vinayaka Missions Annapoorani Matric School             | 11.5971662, 78.0839876 | 30 km/h     | Medium     |
| 6  | university | Vinayaka Missions Vinayaka Missions Research Foundation | 11.5974056, 78.0843617 | 30 km/h     | Medium     |
| 8  | school     | Dist Education Training                                 | 11.6133799, 78.1022409 | 30 km/h     | Medium     |
| 9  | school     | DIET - Salem                                            | 11.6144158, 78.1025078 | 30 km/h     | Medium     |
| 10 | school     | Sri Vidhya Bharati Matric HR Sec School                 | 11.6186069, 78.1103239 | 30 km/h     | Medium     |
| 13 | school     | Sowdeshwari College                                     | 11.6330643, 78.1276021 | 30 km/h     | Medium     |
| 25 | school     | Holy Cross Matriculation Higher Secondary School        | 11.6635657, 78.203532  | 30 km/h     | Medium     |
| 27 | school     | MettupattiThathanur Primary School                      | 11.6827707, 78.2442509 | 30 km/h     | Medium     |
| 30 | school     | Indian Matriculation Higher secondary school            | 12.006299, 78.4531204  | 30 km/h     | Medium     |
| 31 | school     | Indian Matriculation Higher Secondary school            | 12.0061755, 78.4533388 | 30 km/h     | Medium     |
| 33 | school     | Kurinji Matriculation Higher Secondary School           | 12.0156853, 78.4579056 | 30 km/h     | Medium     |
| 34 | college    | Annai Arts and Science College                          | 12.0217961, 78.462795  | 30 km/h     | Medium     |
| 35 | school     | Jayam Vidhyalaya Matriculation Higher Secondary School  | 12.0264103, 78.4656553 | 30 km/h     | Medium     |
| 36 | school     | Harur International School (CBSE)                       | 12.0299967, 78.4693492 | 30 km/h     | Medium     |

Route Photos of Risky Spots



Risk Type: Roundabout  
Risk Level: High



**Speed Limit: 15 KM/Hr**  
**Coordinates: 11.63266, 78.12699**



**Risk Type: Turn**  
**Risk Level: High**  
**Speed Limit: 15 KM/Hr**  
**Coordinates: 11.44029, 77.87544**



**Risk Type: Turn**  
**Risk Level: High**  
**Speed Limit: 15 KM/Hr**  
**Coordinates: 11.44900, 77.87406**

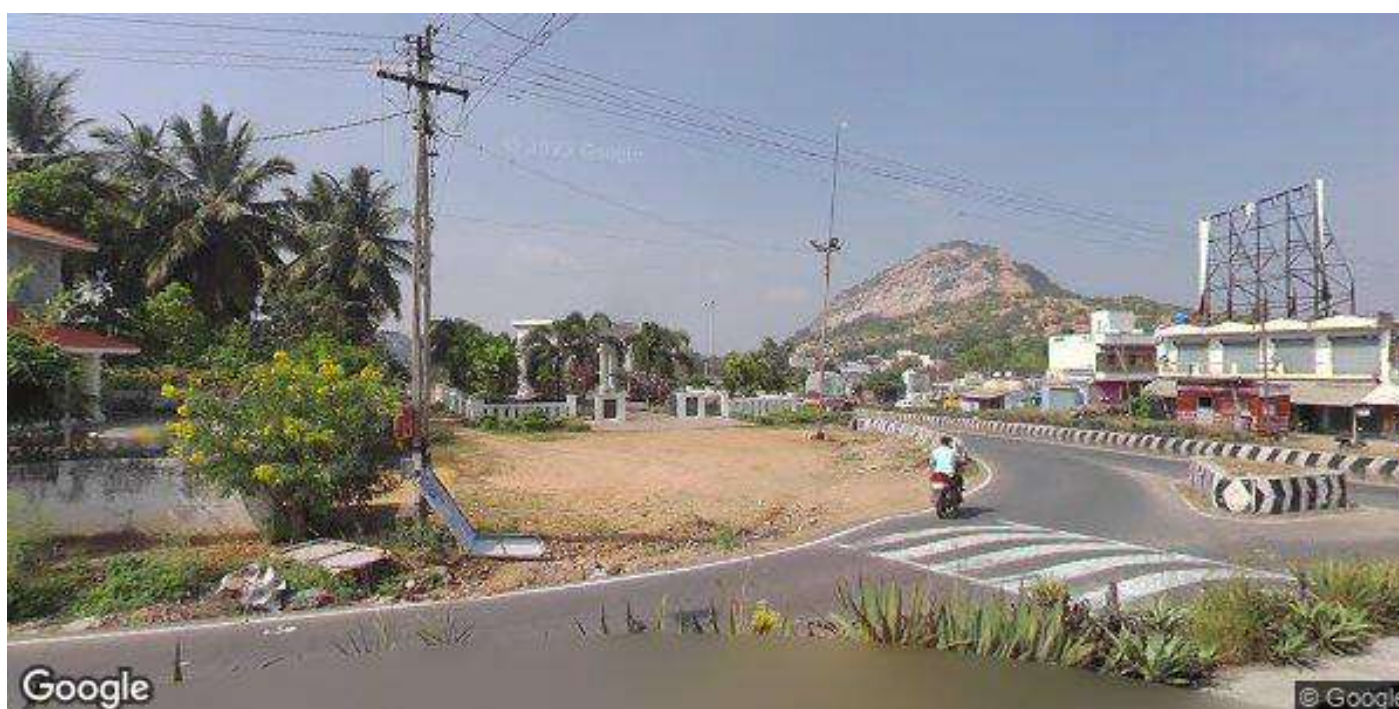




**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45353, 77.85697



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45781, 77.85203



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.46307, 77.84941





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45546, 77.81800



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.45605, 77.81683



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.53573, 77.95794





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.62100, 78.10723



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.62130, 78.10776



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.61235, 78.10438





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.61235, 78.10495



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.62871, 78.12299



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.62788, 78.12219





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.63302, 78.12742

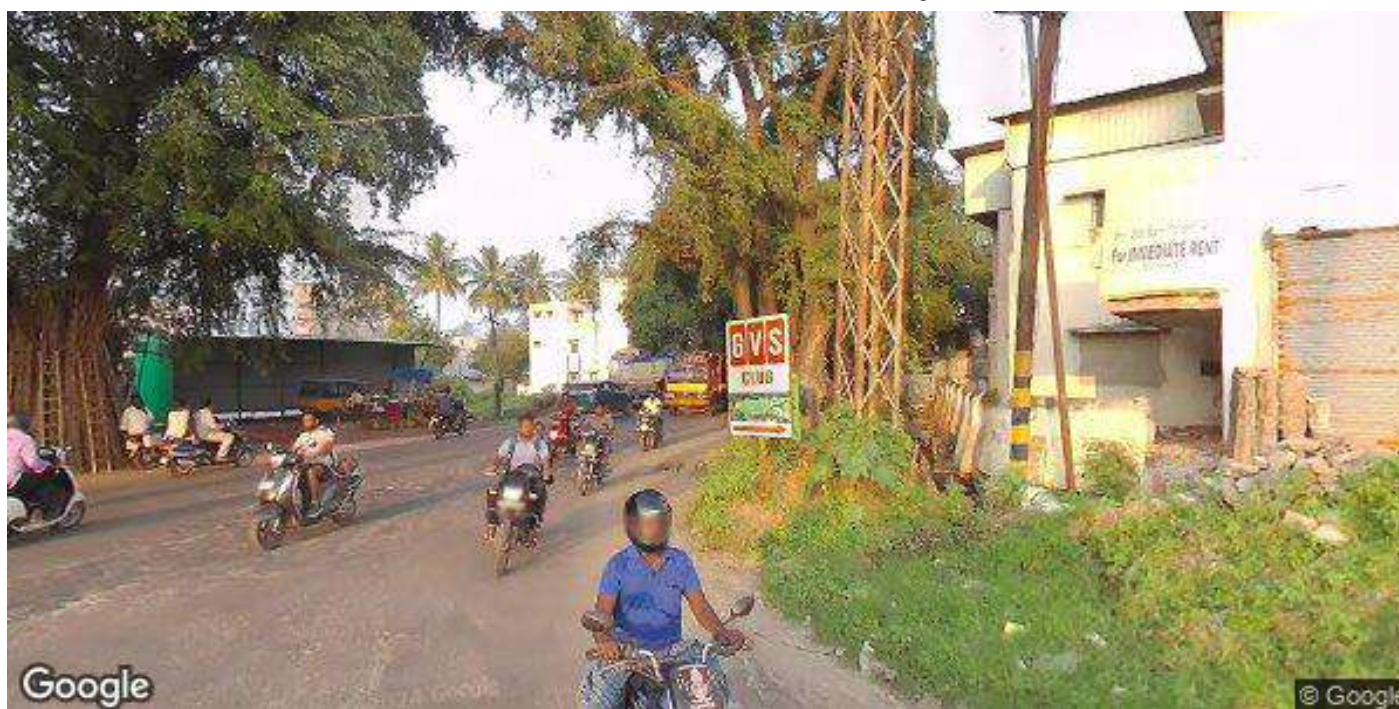


**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.62938, 78.16601



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.65825, 78.20469





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.66366, 78.20649



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.67116, 78.23102



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.67182, 78.23141





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.67500, 78.23962



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.80474, 78.33353



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.81272, 78.34092





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.83514, 78.35305



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.99608, 78.44589



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.04286, 78.48352





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.04845, 78.48385



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.04958, 78.48314



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.06149, 78.47722





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.06932, 78.47655



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.07112, 78.47724



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.11669, 78.49766





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.17764, 78.54800



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.17847, 78.54856



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.18223, 78.54843





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.18614, 78.54962



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.20953, 78.54358



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.21030, 78.54402





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.25956, 78.53301



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.26092, 78.53324



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.26207, 78.53436





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.31095, 78.53818



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.36356, 78.54371



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.36375, 78.54324



